# Frequency Recovery and Readout Effects in Arbitrary-Scale Super-Resolution

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## Abstract

Arbitrary-scale super-resolution (ASISR) predicts pixels at continuous coordinates, but recovery above the low-resolution Nyquist frequency and the role of scale-dependent readout remain difficult to assess. We study these questions with band-wise recovery measurements, covariance-conditioned linear references, and paired cell/footprint ablations. Across three datasets, normalising frequency by the sampling scale reduces variation in all nine evaluated model–dataset groups; the pre-specified collapse tolerance nevertheless passes only six. Evaluated decoders exhibit similar recovery in 10 of 12 low-reference-recovery cells, without establishing a universal decoder limit. Two further training seeds support coarse readout-loss ordering in all four architecture–seed groups. Averaging a full model over training-scale cell inputs predicts loss ordering, but fails function similarity in all four groups and improves on a simple footprint predictor in only two. Classical Wiener mismatch identities supply a conditional interpretation, not a network mechanism. Independent linear experiments corroborate approximate cost ranking, while a frozen 36-case guidance test makes a constant prior choice, trails cross-validation by .112 dB and costs 1.76 times as much. The resulting baseline is an empirical account of scale, frequency recovery and readout effects, with explicit failures of stronger mechanistic and adaptive-selection claims.


## 1 Introduction

Arbitrary-scale super-resolution (ASISR) reconstructs a high-resolution image from a low-resolution image and predicts its value at arbitrary coordinates. Local implicit decoders such as LIIF and LTE combine learned image features with coordinates and a target pixel size [Chen et al., 2021, Lee and Jin, 2022]. A target scale changes both the frequencies directly observed in the input and the pixel footprint used for reconstruction. Aggregate PSNR can obscure which frequency bands improve, and a successful readout ablation does not explain why it improves them.

This paper asks two linked empirical questions: *how does recovery change with frequency relative to the sampling scale, and where does a scale-dependent readout affect that recovery?* We make ASISR frequency recovery and paired readout behaviour the central objects of analysis. Spectral adaptation—changing a representation's preferred frequencies through Fourier features, activation scales or a covariance prior—provides context [Tancik et al., 2020, Sitzmann et al., 2020, Kania et al., 2024]. We do not propose another decoder or claim a general rule for selecting one.

Our measurement is the fraction of band energy recovered, relative to zero reconstruction of the centred signal. We compare trained networks with two specified stationary linear estimators and study frequency in low-resolution Nyquist units, $u = 2s\nu$. A covariance-conditioned Wiener reference distinguishes an assumed prior from the empirical recovery of a trained network. It does not bound all possible estimators on a natural image. Classical mismatch costs explain tendencies in fixed linear channels; their transfer to band-averaged measurements and nonlinear networks is tested separately.

The strongest positive result concerns coarse structure. Frequency normalisation improves scale consistency in nine of nine model–dataset groups. Readout losses generally increase across the three frequency bands, and this ordering replicates in two additional training seeds for two architecture pairs. The stronger accounts fail. Normalisation does not meet the original collapse tolerance in three groups. A predictor based on linear recovery fails the original readout-loss correlation threshold. Scale marginalisation does not reproduce the ablated function, and its advantage over the simpler footprint predictor is architecture-dependent. We preserve these failures rather than replacing the hypotheses after seeing results.

**Contributions and scope.**

1. An ASISR frequency-recovery baseline that separates covariance-conditioned linear references from measured network recovery, and records scale dependence across Set14, BSD100 and Urban100.

2. A replicated paired readout analysis: coarse band ordering is supported in four architecture–seed groups, whereas stronger magnitude, function-equivalence and mechanism claims are evaluated with their counterexamples.

3. An evidence boundary for analytical guidance: classical scalar and matrix identities, independently checked linear cost experiments, and a prospective negative comparison against cross-validation delimit what the ASISR diagnostics do and do not establish.

The formulas are classical tools [Kay, 1993, Rasmussen and Williams, 2006]; the contribution claimed here is the organised empirical evidence and its scope. We distinguish original observations, pre-specified tests, independent reimplementations, saved-record audits and exploratory development. Extended protocols and results accompany the main text in the supplement.

## 2 Related Work

**Spectral control in implicit representations.** SIREN [Sitzmann et al., 2020] sets the frequency scale $\omega_0$ by hand; Fourier features [Tancik et al., 2020] expose the bandwidth $\sigma$ and connect it to the neural tangent kernel [Jacot et al., 2018]; FINER [Liu et al., 2024] widens the reachable frequency set. A recent group selects these parameters from the spectrum of the target signal: FreSh [Kania et al., 2024] matches the spectrum of the untrained model to the target's; WINNER [Chandravamsi et al., 2025] sets Gaussian weight-perturbation scales from the target spectral centroid; SEC [Dądela et al., 2026] retrieves grid-search-derived frequency parameters using centroid-based calibration. WIEN-INR [Ni et al., 2026] uses wavelet-band subnetworks and a finest-detail enhancement module. These studies include signal fitting, and FreSh and WINNER also evaluate inverse tasks. We focus on observation- and noise-conditioned linear references, mismatch costs, and the empirical reliability of guidance under degraded observations. Spectral bias [Rahaman et al., 2019] explains what an untuned network prefers; our covariance-conditioned reference describes recovery under a specified signal law and observation model.

**Arbitrary-scale super-resolution.** LIIF [Chen et al., 2021] decodes a local latent at continuous coordinates with a cell (footprint) input; LTE [Lee and Jin, 2022] adds a local dominant frequency estimator; MetaSR [Hu et al., 2019], SRNO [Wei and Zhang, 2023], CiaoSR [Cao et al., 2023] and HIIF [Jiang et al., 2025] refine the decoder; the SC-INR research implementation examined here replaces the cell input with an analytic footprint (sinc) readout and includes scale-consistency components; their benefits are evaluated separately from the footprint. Gaussian-splatting decoders [Chen et al., 2025] and heat-field decoders with analytic anti-aliasing [Becker et al., 2025] form a newer family. A controlled re-training study [Nasir et al., 2026] examines the influence of decoder architectures and training recipes. Low-pass behaviour of SR networks has been measured with a global scalar [Deng et al., 2024]; we measure recovery per band defined by the sampling theorem and set it against a covariance-conditioned linear reference. We examine these empirical similarities and their limitations.

**Multi-frame and sparse reconstruction.** SuperF [Jyhne et al., 2025] fits a Fourier-feature network to multiple shifted low-resolution frames and tunes $\sigma$; Deep Image Prior [Ulyanov et al., 2018] reconstructs without training data; kriging/GP regression [Rasmussen and Williams, 2006] inserts the spectrum directly as a prior. These tasks serve as linear and inverse-problem checks of the diagnostic, separate from the ASISR readout analysis.

**Classical theory and per-band diagnostics.** LMMSE/Wiener estimation and aliasing analysis of sampling are textbook [Kay, 1993]. Per-frequency agreement measures such as Fourier ring correlation in microscopy [van Heel and Schatz, 2005] and scale-decomposed skill scores in forecast verification are close relatives of our per-band recovery score. We use the score to compare scale-dependent ASISR behaviour, rather than claiming a new information measure or a general spectral-selection theorem.

## 3 Measurement and Conditional Linear References

**Setting.** A signal $x$ lives on a continuous domain $\Omega \subset \mathbb{R}^d$. We observe $y = Ax + n$ with a known linear operator $A$ (blur and decimation, sparse sampling, or the footprint integrals of several shifted frames) and $n \sim \mathcal{N}(0, \sigma_n^2 I)$. A continuous representation $f_\theta$ is fitted by minimising $\|Af_\theta - y\|^2$ (possibly with regularisation or early stopping) and queried anywhere.

**Proposition 1** (Fourier features are a GP prior; Bochner)**.** *Let* $f(u) = \sum_{j=1}^{m} w_j \varphi_j(u)$ *with* $\varphi_j(u) = \sqrt{2/m}\cos(2\pi b_j^\top u + c_j)$, $b_j \sim p(b)$, $c_j \sim \mathcal{U}[0, 2\pi)$ *and* $w \sim \mathcal{N}(0, s^2 I)$. *As* $m \to \infty$, $f$ *is a zero-mean stationary Gaussian process with covariance* $k(h) = s^2\, \mathbb{E}_b[\cos(2\pi b^\top h)]$ *and spectral density* $S(\nu) = s^2[p(\nu) + p(-\nu)]/2$, *equal to* $s^2 p(\nu)$ *for symmetric* $p$.

Hence a ridge read-out of Fourier features is the GP posterior mean under prior spectrum $S$; with $p = \mathcal{N}(0, \sigma^2 I)$, $S(\nu) \propto \exp(-\|\nu\|^2/2\sigma^2)$. Tuning $\sigma$ (SuperF) or choosing frequency parameters from a target spectrum (FreSh, WINNER, SEC) controls frequency preference; the exact GP interpretation applies to the linear feature readout within a restricted family; a GP with a fitted or known spectrum picks it directly. For an MLP on Fourier features trained by gradient descent, the lazy-training limit gives kernel regression with a stationary NTK modulated by $p$ [Tancik et al., 2020, Jacot et al., 2018], under additional architectural and optimisation assumptions.

General trained MLPs and grid fields do not thereby have known stationary spectral priors; we compare their behaviour empirically (§A.4).

**Proposition 2** (Covariance-conditioned linear optimality)**.** *For zero-mean or correctly centred signal and noise, let $x$ have covariance $C_P$, uncorrelated noise covariance $\sigma_n^2 I$, and positive-definite $C_y$. Every linear estimator $\hat{x} = Gy$ satisfies $\mathbb{E}\|x - \hat{x}\|^2 \geq \mathbb{E}\|x - \hat{x}_{\mathrm{LMMSE}}(P)\|^2$, with $\hat{x}_{\mathrm{LMMSE}}(P) = C_P A^\top (A C_P A^\top + \sigma_n^2 I)^{-1} y$.*

This optimum holds in expectation under the specified covariance and noise. Second moments and zero signal–noise cross-covariance suffice for linear optimality; joint Gaussianity additionally identifies the conditional mean with MMSE. A generic spectrum does not define a distribution-free ceiling on real images, and an image periodogram is an oracle reference rather than an attained optimality certificate. For any fixed linear $H$ and evaluation map $W$, with $C_y = AC_PA^\top + C_n$ positive definite,

$$\begin{aligned}&\mathbb{E}\|W(x - Hy)\|^2 - \mathbb{E}\|W(x - H_P y)\|^2 \\ &= \mathrm{tr}[W(H - H_P)C_y(H - H_P)^\top W^\top] \geq 0,\end{aligned}$$

where $H_P = C_P A^\top C_y^{-1}$. The same covariance-conditioned optimum minimises total and projected linear risk.

**Definition 1** (Per-band linear recoverability)**.** For a band projector $\Pi_b$,

$$R_b^{\mathrm{lin}}(A, \sigma_n, P) = 1 - \frac{\mathbb{E}\|\Pi_b(x - \hat{x}_{\mathrm{LMMSE}}(P))\|^2}{\mathbb{E}\|\Pi_b x\|^2}.$$

$R_b^{\mathrm{lin}}$ depends only on $A$, $\sigma_n$ and $P$. With a fully specified generic natural-image prior $P \propto 1/f^2$ it requires no target ground truth and no training; we compute it by sampling from the prior, simulating the observation and applying the LMMSE, with explicit reference-sampling and solve costs. Its amplitude is an additional prior assumption. When a trained or otherwise non-linear estimator is evaluated on real images, we report the same quantity measured on data, $R_b = 1 - \|\Pi_b(x - \hat{x})\|^2 / \|\Pi_b(x - \bar{x})\|^2$ using the colour aggregation specified by each protocol; for a trained network we write $R_b^{\mathrm{net}}$.

**Bands.** Image bands are square annuli in HR pixel units, indexed by $\max(|\nu_x|, |\nu_y|)$, $b_1 = [0, \frac{1}{8})$, $b_2 = [\frac{1}{8}, \frac{1}{4})$, $b_3 = [\frac{1}{4}, \frac{1}{2}]$ cycles/pixel. At $\times 4$, $b_1$ lies below the LR Nyquist limit and $b_2$, $b_3$ lie one and two octaves above it.

**Closed form for decimation.** For ideal periodic $A =$ blur $K$ followed by decimation by $s$, each LR frequency $\omega$ collects $s^d$ HR frequencies $\nu_m = \omega + m/s$, and

$$R(\nu_0) = \frac{|\hat{K}(\nu_0)|^2 P(\nu_0)}{\sum_m |\hat{K}(\nu_m)|^2 P(\nu_m) + N_\omega}. \tag{1}$$

The sum in the denominator is aliasing. The historical simulations show little change over the particular tested noise range (Table 17); this is not a general noise-insensitivity claim.

Here $N_\omega$ is noise power under a compatible Fourier normalisation. Changing other powers within the alias set changes the effective noise; the scalar mismatch identity assumes that effective noise is fixed. Finite crops, boundary interpolation, rotations and sparse sampling normally require the matrix risk expression. F1/G1 use HR256 and 73 images; L1/L2 use HR240 and 74, with their own window and pooling conventions. Empirical recovery can be negative and need not equal reference recovery.

Zero-noise singular cases require a pseudoinverse or explicitly stated regularisation. A numerical noise floor changes the estimator; its value must not replace the physical observation-noise covariance in risk evaluation.

**Energy pooling and scale normalisation.** Write $S_{ib}$ and $E_{ib}$ for signal and reconstruction-error energy in image $i$, band $b$, under a stated crop, colour conversion and window. The pooled measurement is

$$R_b^{\mathrm{emp}} = 1 - \frac{\sum_i E_{ib}}{\sum_i S_{ib}}. \tag{2}$$

It is not an unweighted mean of per-image recoveries. Negative recovery means worse squared error than a centred zero reconstruction. For L1, frequency is the Chebyshev radius $\nu = \max(|k_x|, |k_y|)/240$. Seven bins in $u = 2s\nu$ place the LR Nyquist frequency at $u = 1$. For each bin we take the range across $s \in \{2, 3, 4\}$, then the median across bins:

$$D_{\mathrm{in}} = \mathrm{median}_b \left[ \max_s R_b(s) - \min_s R_b(s) \right]. \tag{3}$$

The absolute-frequency control fixes bins at the $s = 3$ reference, $\nu = u/6$. A smaller $D_{\mathrm{in}}$ is greater consistency in this measurement, not proof that two reconstructed functions are identical.

# 4 Classical Mismatch Costs and Diagnostic Limits

For a centred scalar observation with fixed effective noise variance $q$ and signal variance $p$, the matched Wiener gain is $R = p/(p + q)$. An estimator assuming variance $\kappa p$ uses $g = \kappa R/(1 - R + \kappa R)$. The normalised excess error is a classical identity:

**Proposition 3** (Scalar mismatch cost)**.**

$$\Delta(R, \kappa) = \frac{R(1 - R)^2(\kappa - 1)^2}{(1 - R + \kappa R)^2}. \tag{4}$$

The proof in Appendix B separates this identity from its experimental approximations. In absolute units, for true power $P$, assumed power $Q$ and unchanged effective noise $N$,

$$P\Delta = \frac{N^2(Q-P)^2}{(P+N)(Q+N)^2}. \tag{5}$$

Under-estimation has limit $\Delta \to R$ as $\kappa \to 0$, whereas over-estimation has limit $(1-R)^2/R$ as $\kappa \to \infty$. The latter exceeds signal energy at $R < (3-\sqrt{5})/2$; total error exceeds zero reconstruction when $\Delta > R$. Neither statement defines a universal safe bandwidth.

**Corollary 1** (Reciprocal mismatch)**.** *For* $0 < R < 1$ *and* $\kappa > 1$, $\Delta(R,\kappa) > \Delta(R, 1/\kappa)$ *if and only if* $R < 1/2$.

The effective noise is held fixed, including any aliased components represented by it. Changing several powers in an alias set or using a non-diagonal sampling operator need not preserve this scalar reduction.

**Reference labels, not natural boundaries.** Historical protocols partition reference recovery at .1 and .8: A denotes high reference recovery, B intermediate and C low. These fixed labels permit the original test criteria to be audited. Their sensitivity has not been established, so we do not recommend them as deployment thresholds. With a fixed finite $\kappa$, the scalar excess vanishes as $R \to 0$ or 1, but the low-$R$ limit is not uniform over large mismatch factors. No reference label identifies the causal bottleneck of a trained decoder.

**Band approximation.** A diagnostic sum

$$C(Q) = \sum_b E_b \, \Delta\big(R_b, \kappa_b(Q)\big) \tag{6}$$

substitutes measured band energies, reference recoveries and ratios of spectrum integrals. It is not the exact risk for a general operator. Its ranking accuracy and winner selection are distinct tests (§6). Averaging $R$ and $\kappa$ before the nonlinear formula can either over- or under-estimate the true sum. For a fixed linear estimator the matrix expression in §3 instead evaluates the specified covariance law directly.

**No network mechanism follows.** A trained ASISR decoder is not a fixed Wiener estimator with known $\kappa$. Its measured $R_b^{\mathrm{net}}$ is a truth-dependent outcome. Values near zero or one do not force a small readout-ablation loss: removing a component may improve a poorly recovered band or damage a perfectly recovered one. The loss ordering and scale-marginalisation tests below therefore require empirical evidence independent of the scalar identity.

# 5 ASISR Frequency and Readout Baseline

**Protocol and evidence units.** We retain the recovered hypotheses and thresholds. Table 1 distinguishes crops, image counts and measurement conventions. Cells share images, scales, bands and checkpoints; they are not independent experimental replications. A model–seed group is the unit for the four-group replication criteria. All original results and failures remain in the supplement. No new training or target-dependent tuning was used for this revision.

## 5.1 Recovery changes with scale

LIIF, LTE and SC-INR variants use the inherited EDSR-baseline/DIV2K training setup, 1000 epochs and scales ×1–×4. The early E3 comparison includes SC-INR-eq and SC-NoSinc but is not their matched ablation; F1/G1 use the correct paired models. Network equality or architectural saturation is not assumed.

At ×4, both upper bands have low recovery under the generic stationary reference. The trained decoders recover more than the evaluated oracle-periodogram linear reference, with similar values within each dataset (Table 2). Across the additional scales ×2/3/8, the historical reference ranks measured generic-linear recovery with Spearman .952 over 27 cells; generic recovery decreases monotonically in all nine band–dataset pairs. Under the frozen reference labels, all four networks exceed the oracle reference in 12/12 C cells, but their range is at most .03 in 10/12 and at most .02 in 8/12. The best network is within .05 of the oracle in 9/9 A cells. These are finite descriptive comparisons, not optimality certificates.

**What the reference actually measures.** Original E3 prior simulation injects $\sigma = .002$ noise. Its natural-image evaluation adds none, yet uses GP variance $\nu = 10^{-5}$ as regularisation; generic amplitude is scaled with HR truth variance. LR clipping and DC handling also differ between the reference and real-image paths. Thus the comparison supports scale ordering under those implementations, not a matched-noise, observation-only bound. The new record audit reproduces all 27 cells and criteria; missing predictions, band energies and true CG residuals prevent an independent pixel-level or zero-noise sensitivity recomputation. These limits are carried into the conclusions rather than repaired by changing the meaning of $\nu$.

## 5.2 Nyquist normalisation improves consistency, not exact collapse

The L1 hypothesis tests $R(u)$ with $u = 2s\nu$ using Eq. (3). Binning by $u$ reduces scale variation in every model–dataset group: $D_{\mathrm{in}} = .013$–$.061$, versus .31–.46 for the

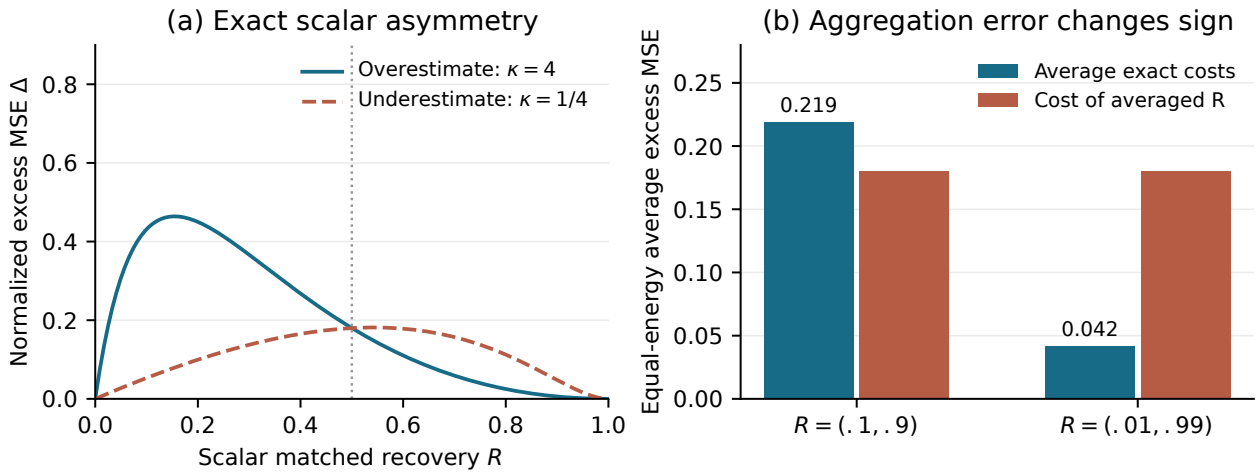



Figure 1: Classical scalar cost and aggregation limits. Reciprocal mismatches cross at $R = .5$ under fixed effective noise. Equal-energy modes with the same mean $R = .5$ and $\kappa = 4$ can give aggregation errors of either sign. These calculations do not identify a nonlinear network's effective prior.

Table 1: ASISR protocol alignment. Each count is Set14/BSD100/Urban100, using the first eligible images from the recovered protocol. ×8 is excluded from the primary 27-cell G1/L2 tests.

| Test | HR crop / images | Evaluated scales | Colour/pooling | Role |
|---|---|---|---|---|
| E3 | 256 / 13,30,30 | 2,3,8 (+historical 4) | RGB mean; band energy ratio | reference and network comparison |
| L1 | 240 / 14,30,30 | 2,3,4 (+6,8,12) | pooled signal/error energies | scale-normalised consistency |
| F1/G1 | 256 / 13,30,30 | 2,3,4 | RGB mean; energy pooling | paired readout loss |
| L2 | 240 / 14,30,30 | 2,3,4 (+8) | pooled RGB energies | marginalisation prediction |

Table 2: ×4 empirical recovery. Network range includes LIIF, LTE, SC-INR-eq and SC-NoSinc. The oracle uses the image spectrum; it is unavailable from LR data alone.

| Dataset | Band | Generic | Oracle | Network range |
|---|---|---|---|---|
| Set14 | $b_2$ | .16 | .25 | .56–.57 |
| | $b_3$ | .09 | .09 | .18–.20 |
| BSD100 | $b_2$ | .13 | .22 | .38–.38 |
| | $b_3$ | .06 | .06 | .11–.13 |
| Urban100 | $b_2$ | .11 | .27 | .50–.51 |
| | $b_3$ | .02 | .03 | .13–.15 |

Table 3: L1 $D_{\text{in}}$ from saved pooled energies. Set14 values exceed the frozen .05 tolerance.

| Model | Set14 | BSD100 | Urban100 |
|---|---|---|---|
| LIIF | 0.0586 | 0.0154 | 0.0165 |
| LTE | 0.0548 | 0.0149 | 0.0190 |
| SC-INR | 0.0605 | 0.0128 | 0.0147 |

fixed absolute-frequency bins. However, the pre-specified requirement $D_{\text{in}} \leq .05$ in at least eight of nine groups fails: only six pass (Table 3). The three Set14 failures exceed the linear control and the .003–.004 reported seed-noise range. This supports a useful normalisation, not a universal collapse law.

**Content sensitivity.** A post-hoc leave-one-image-out check identifies barbara as the largest contributor to the Set14 discrepancy. Removing it reduces $D_{\text{in}}$ to .0361/.0298/.0300 for LIIF/LTE/SC. For comparison, 200 fixed-seed samples of 14 images from the first 30 BSD100 images have respective maxima .0392/.0378/.0388. This diagnoses sensitivity of energy pooling to image content; it does not establish JPEG artefacts or a capacity mechanism. We retain the original failed cohort and criterion. Extrapolation and public-model checks likewise remain limited: the original outside-training-range criterion and public-model aggregate each pass only six of nine groups.

### 5.3 Paired readout losses and coarse band ordering

The readout comparisons pair LTE with LTE-NoCell and SC-INR with SC-INR-NoSinc under the recovered matched training setup. Define

$$L_b = R_b^{\text{full}} - R_b^{\text{ablated}}. \tag{7}$$

Positive $L_b$ means the full readout improves recovery; negative values are retained. A paired training ablation measures the change between trained models, not the

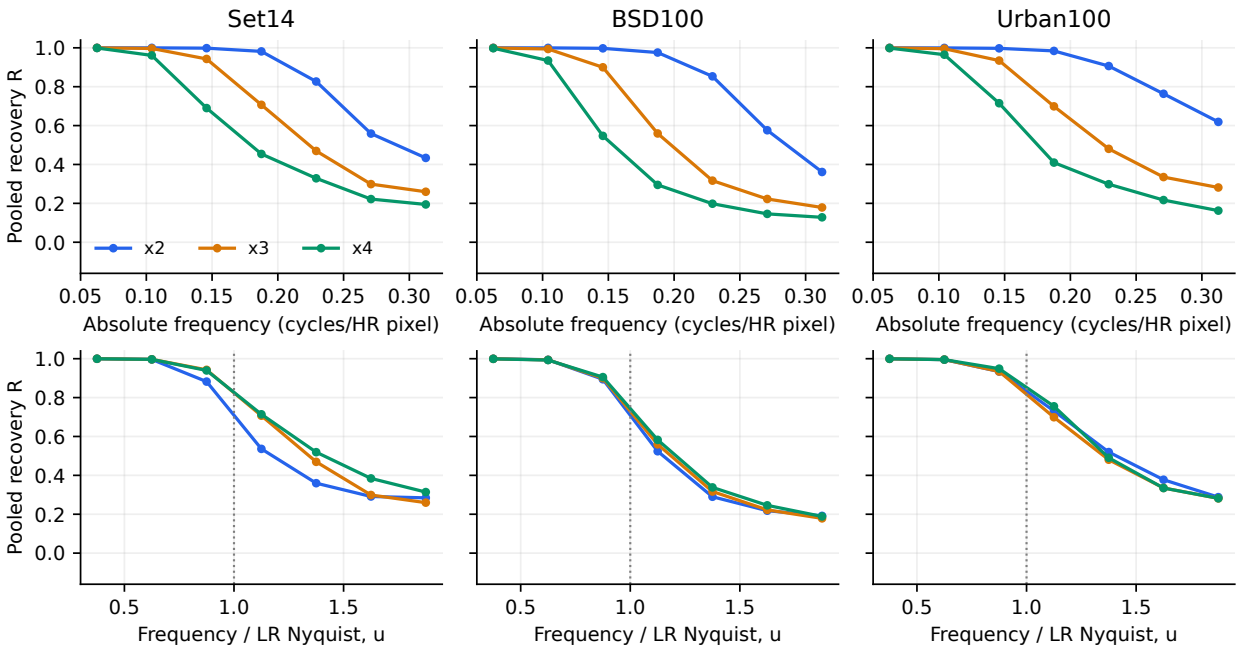



Figure 2: Scale-dependent recovery of SC-INR (seed1), from the saved L1 energy arrays. Top: fixed absolute-frequency bins. Bottom: the same measurement in LR Nyquist units $u = 2s\nu$; the vertical line marks $u = 1$. Curves use pooled energy ratios, not a mean of image-wise recoveries. The axes describe evaluated error and need not stay in $[0, 1]$. Normalisation substantially reduces scale differences; Table 3 retains the Set14 threshold failure and all three-model results.

effect of deleting a component while holding all learnt weights fixed.

The original F1 small-mismatch predictor $g(R) = R(1-R)^2$ fails the .5 correlation threshold for both architectures: .411/.400 with linear recovery, .490/.492 with network recovery. Cross-architecture loss ranking is high (.985), but magnitudes differ. For the correct SC pair, mean losses over $b_1/b_2/b_3$ are $-.0000142/+.0001368/+.0080551$; A-labelled cells average $-.0006091$, and $\times 3$ $b_2$ averages $+.0010338$ (LTE: $+.0050$). Some SC cells improve after ablation, including Set14 $\times 2$ $b_3$ ($-.03788$). High rank correlation therefore does not imply equal benefits or a common quantitative mechanism.

G1 tests a simpler predictor: the box-footprint versus point-sampling quantity $\epsilon(\nu) = 1 - \mathrm{sinc}(\nu_x)\,\mathrm{sinc}(\nu_y)$ gives fixed squared-band scores $(.00006, .00552, .07235)$. It was proposed after seed1 and frozen before seed2/3 evaluation. The full original protocol has been recovered and hash-verified. Table 4 reports all four confirmatory groups.

The original ordering threshold passes in four of four groups; $L_3 > L_2$ occurs in 34/36 settings. The additional recovery factor supplies no improvement under the frozen criterion. This supports coarse frequency ordering across these architecture–seed groups. It does not validate a gain-mismatch explanation, nor a precise loss amplitude. Saved pooled statistics and coverage are independently checked; G1 did not save per-image signal/error energy weights, so full independent repooling is unavailable.

Table 4: G1: 27 cells per group. Average ranks handle the predictor's three tied band values. Adding $h(R) = R(1-2R)^2$ reduces correlation in every group.

| Group | $\rho(\epsilon_b^2, L)$ | $\rho(\epsilon_b^2 h(R), L)$ | $L_3 > L_2$ |
|---|---|---|---|
| LTE seed2 | .577 | .532 | 8/9 |
| SC seed2 | .943 | .703 | 9/9 |
| LTE seed3 | .676 | .643 | 8/9 |
| SC seed3 | .885 | .804 | 9/9 |

## 5.4 Scale marginalisation predicts ordering but not the ablated function

L2 asks whether a network without a scale readout behaves like the full model averaged over the training-scale inputs. At fixed coordinates and image features, we average predictions over 31 cell inputs corresponding to $s' \in \{1.0, 1.1, \ldots, 4.0\}$, obtaining $R_{\mathrm{marg}}$. The proposed loss is $L_{\mathrm{pred}} = R_{\mathrm{full}} - R_{\mathrm{marg}}$. The primary test covers the 27 $\times 2/3/4$ cells; log-spaced queries and $\times 8$ are secondary. This is a post-training diagnostic using HR truth, not a deployable LR-only rule.

Ordering passes in four groups and the magnitude interval $[.5, 2]$ in three. Function similarity never reaches .8. Superiority over the simple footprint score by at least .05 occurs in two groups, below the required three; both are LTE. The cell-invariant ablation control passes in all

Table 5: Complete L2 seed2/3 replication. Ratio is the median $L_{\text{pred}}/L$ over $L > .005$. Similarity is the fraction of image–scale pairs where the ablated output is closer to the marginalised than the full output.

| Group | $\rho$ | Ratio | Similarity | $\rho - \rho_\epsilon$ |
|---|---|---|---|---|
| LTE seed2 | .712 | .823 | .770 | +.130 |
| SC seed2 | .755 | .581 | .689 | −.142 |
| LTE seed3 | .954 | .833 | .748 | +.331 |
| SC seed3 | .700 | .337 | .653 | −.173 |

four groups with maximum difference zero. Independent reconstruction from per-image signal/error energies reproduces the pooled values. The result supports empirical prediction of ordering, while rejecting function equivalence and a uniformly better account across architectures. G1 and L2 differ in crop, image count and colour aggregation, so their correlations are compared within their own protocols.

# 6 Checks and Boundaries Beyond ASISR

The following completed experiments constrain interpretation of the ASISR baseline. Their full tables and protocols are supplementary; none supplies a nonlinear readout theorem or an adaptive INR selector.

**Linear mismatch checks.** E2 uses 12 DIV2K images, five tasks, three bands and four mismatch factors. An independent implementation reproduces all 60 aggregated cells: cost ranking .903, generic-reference ranking .797, median absolute error .0351, but maximum error 1.8895. Low/high-recovery asymmetry counts are 4/4 and 6/7. Windowing was added after a one-image pilot, so the metric was not wholly specified before seeing data. Approximate cost ranking is stronger than its numerical accuracy; the original and rerun values remain distinct.

**Oracle diagnosis versus feasible selection.** In sparse reconstruction the true-spectrum linear reference exceeds DIP at all four sampling rates, while the generic linear estimator trails DIP at all four. This is an oracle gap, not evidence that a deployable generic GP beats DIP. In multi-frame E1, two of four original switching predictions fail. D11's truth-derived band cost ranks 24 arm–noise points with $\rho = .856$ (10k image-bootstrap interval $[.741, .918]$), but picks the best non-oracle arm at only one of three noise levels. Rank correlation does not establish reliable winner selection.

**Frozen prospective guidance.** G2 uses an independent 32-image calibration and a complete, frozen 12-image by three-noise cohort, choosing among eight fixed linear GP spectra before reading target truth. Analysis improves the specified power2 default by 1.105 dB (image-cluster 95% interval $[.726, 1.504]$), but all 36 choices are power2.5. It trails four-fold frame CV by .112 dB (interval $[-.313, -.002]$) and costs 8466.8 versus 4821.7 seconds. A fixed power2.5 reconstruction would give identical outputs; its 194.6-second retrospective cost excludes the work needed to choose that static prior. It is not a new confirmatory baseline or a selection speedup. The test does not establish target-specific adaptation, a cost advantage or INR architecture selection.

**Exploratory evidence retained separately.** Same-observation linear/FF controls on two already-seen images include reverse rankings. A common 32-image empirical evaluation law selects the same strong static prior; it does not isolate a nonlinear mechanism or demonstrate adaptation. An 8-point Gaussian example and the associated finite-sample confidence-set study concern fresh-draw population risk, not single-image safety; they remain classical illustrative material. EEG supplies only 200 calibration segments, with the 760-segment test archive missing. These results motivate scope restrictions rather than cross-modal or new-theory contributions.

# 7 Limitations

**Reference and numerical scope.** Linear optimality is conditional on the specified covariance, centring, operator and noise. The image-periodogram oracle is truth-dependent. E3's assumed variance, clipping, amplitude scaling and missing pixel/energy/residual archives limit its use to the recovered descriptive comparison. Exact matrix identities do not certify an approximate solver or a clipped output. E1 also has a native forward/inverse discretisation mismatch.

**Empirical generalisation.** ASISR results cover a limited set of decoders, training recipes, datasets and three seeds. Bands and cells share images and are dependent. Similar recovery is not a Bayes limit or proof that more capable decoders cannot improve. Nyquist normalisation improves consistency but fails the original strict threshold in Set14; content sensitivity and readout losses remain empirically observed, not causal explanations.

**Mechanism and guidance.** Classical scalar identities require fixed effective noise and cannot be substituted into a nonlinear network without further assumptions. G1 supports only coarse ordering; L2 fails function similarity and general superiority over a simpler predictor. Threshold sensitivity at .1/.8 remains untested. G2 covers one operator, one calibration and fixed noise draws; its constant selection and worse comparison to CV leave

Table 6: Claim hierarchy retained in the baseline. Passing a weaker test does not upgrade a failed stronger one.

| Question | Complete evidence | Supported conclusion |
|---|---|---|
| Does normalisation reduce scale variation? | L1 improves 9/9; strict tolerance passes 6/9 | Useful normalisation; no universal collapse |
| Can a fixed band score order readout losses? | G1 ranks 4/4; adding recovery factor improves 0/4 | Coarse ordering; no gain mechanism |
| Does scale averaging reproduce the ablation? | L2 ranking 4/4, magnitude 3/4; similarity 0/4 | Correlational prediction; no function equivalence |
| Is L2 better than the simpler predictor? | Improvement in 2/4, below required 3/4 | No general superiority |
| Does calibrated analysis adapt to each target? | G2 constant choice in 36/36; worse accuracy/cost than CV | Default improvement; no adaptive advantage |

adaptive and computational benefits unsupported. No learned INR model-choice rule is established.

**Preserved auxiliary evidence.** Supplementary controls use already-seen development images and single seeds. Their reverse rankings and static-baseline equivalence are retained. The Gaussian ambiguity example depends on the allowed estimator class; a restricted two-candidate regret bound is not an all-estimator reconstruction lower bound. Classical covariance-estimation and design tools are not claimed as new theorems. Calibration-only EEG does not supply independent cross-modal validation.

# 8 Conclusion

The paper baseline concerns scale-dependent frequency recovery and readout effects in ASISR. Nyquist normalisation reduces scale variation, and coarse readout-loss ordering replicates across two architecture pairs and two additional seeds. Stronger collapse, function-equivalence, quantitative mechanism and adaptive-selection claims remain unsupported by their respective tests. Covariance-conditioned linear references and classical mismatch costs help organise these observations, with their assumptions and numerical limits stated separately. Further innovation should be assessed against this preserved baseline rather than inferred from additional correlations or oracle comparisons.

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Table 7: Original E2 reported criteria (60 cells, 12-image means) .

| Criterion | Threshold | Result |
|---|---|---|
| a. Spearman(band-avg. prediction, measured) | $\geq 0.8$ | **0.90** |
| b. $R < 0.3$: over-est. > under-est. | $\geq 80\%$ | 4/4 |
| b. $R > 0.7$: under-est. > over-est. | $\geq 80\%$ | 6/7 |
| c. median \|pred.−meas.\| | $\leq 0.05$ | 0.038 |
| d. Spearman(a-priori prediction, measured) | $\geq 0.7$ | 0.79 |

Table 8: Original E2 measured / band-averaged predicted cost $\Delta$ (fraction of band energy) .

| Task, band | $R$ | $\kappa$=$\frac{1}{16}$ | $\frac{1}{4}$ | 4 | 16 |
|---|---|---|---|---|---|
| SI×2 $b_2$ | .80 | .230/.402 | .047/.101 | .067/.031 | .165/.056 |
| SI×2 $b_3$ | .11 | .084/.097 | .044/.058 | .194/.380 | **1.370**/2.648 |
| SI×4 $b_2$ | .085 | .060/.074 | .029/.045 | .150/.336 | **0.933**/2.590 |
| Sparse 10% $b_2$ | .37 | .197/.291 | .072/.144 | .110/.302 | .586/.976 |
| Sparse 25% $b_2$ | .70 | .226/.430 | .068/.139 | .094/.074 | .473/.149 |
| Sparse 50% $b_2$ | .90 | .148/.264 | .036/.046 | .024/.009 | .064/.015 |

# Supplementary material

Extended ASISR results and auxiliary experiments are followed by classical derivations, reproducibility limits and exploratory controls.

# A Extended experiments and recovered protocols

**Protocol.** Confirmatory thresholds follow the recovered pre-registrations. Freezing is audited experiment by experiment: E2 windowing changed after a one-image pilot, G1 predictors were proposed after seed1, and exploratory analyses are not promoted to confirmation. Images are from DIV2K validation (0801–0812) unless stated; $R_b$ uses each recovered protocol: F1/G1/E3 average RGB to grayscale, L2 pools RGB channel energies, and E1/D11 retain RGB metrics. Exploratory analyses are labelled as such.

## A.1 The mismatch cost is predictable (E2)

**Setup.** 12 DIV2K images, five tasks (single image ×2, ×4; sparse pixels 10%, 25%, 50%). In each of the three bands we multiply the true spectrum by $\kappa \in \{1/16, 1/4, 4, 16\}$, reconstruct with a GP (linear), and measure the lost recoverability in that band, giving 60 cells . Band energies are computed after a Hann window on both signal and error; the window was not specified in the pre-registration and was added after a one-image run showed boundary leakage (single-image ×4 $b_3$ measured $R = 0.56$ at 0.004% transmittance). After the fix, single-image ×4 $b_2$ measures $R = 0.085$ against an a-priori 0.08 . Only the metric changed, not hypotheses or criteria.

**Independent reimplementation (E2r).** A separate implementation reproduces all 60 aggregated cells from 12 images and five tasks. Its post-hoc cost ranking is .902528, reference-prior ranking .797188 and median absolute error .035099 (maximum 1.8895). Asymmetry counts are 4/4 and 6/7; all four frozen thresholds pass. These are distinct rerun numbers, not replacements for individual original cells in Tables 7 and 8. The high maximum error limits quantitative interpretation despite the passed median threshold .

**Results.** All four pre-registered criteria hold (Table 7). For example, single-image ×4 $b_2$ has under-estimation cost .060 and over-estimation cost .933 for reciprocal factors 1/16 and 16; the latter exceeds the matched recovery .085 and hence is worse than zero band reconstruction (Table 8). In high-$R$ bands the direction reverses. In the reported near-zero-recovery cell (single image ×4 $b_3$), both evaluated finite mismatch factors incur small costs; this is not uniform over arbitrarily large factors . The band-averaged prediction over-estimates large-$\kappa$ costs in the reported examples; this direction is not general. For decimation, evaluating Eq. (4) per frequency with Eq. (1) and energy-weighting reduces approximation error in the reported exploratory calculation (exploratory: Spearman 0.99, median error 0.004; e.g. SI×4 $b_2$, $\kappa = 4$: predicted 0.151, measured 0.150) . The single counter-example to the asymmetry is sparse 10% $b_1$ ($R = 0.92$; under 0.072 vs. over 0.110): random sampling spreads aliasing across bands, outside the scalar model .

## A.2 Low-reference-recovery bands: similarities among evaluated ASISR decoders (M2, E3)

**Setup.** We compare four trained decoders that share the EDSR-baseline encoder and the LIIF training protocol (DIV2K, 1000 epochs, scale range ×1–×4, seed 1): LIIF, LTE, SC-INR-eq and SC-INR without the sinc footprint readout. This early comparison is not a matched SC-INR/NoSinc ablation; the latter is evaluated separately in F1/G1 against two linear references, LMMSE with the generic spectrum and LMMSE with the image's own (oracle) spectrum, on $256^2$ centre crops of the first 30 images of Set14, BSD100 and Urban100 .

**×4.** In both regime-C bands the trained networks recover more than the evaluated oracle-periodogram linear reference: 0.38–0.57 vs. 0.22–0.27 in $b_2$ (1.7–2.3×) and 0.13–0.20 vs. 0.03–0.09 in $b_3$ (2–6×) (Table 9). This comparison is consistent with benefits from a learned prior, but does not prove that all linear models or all alternative

Table 9: Single image ×4, $R_b$ measured on data . Regime by $R_b^{\text{lin}}$: $b_2$, $b_3$ are C. gen./orac.: LMMSE with generic/oracle spectrum; SC-eq/SC-ns: SC-INR-eq / SC-INR-nosinc.

| Set | band | gen. | orac. | LIIF | LTE | SC-eq | SC-ns |
|---|---|---|---|---|---|---|---|
| Set14 | $b_2$ | .16 | .25 | .56 | .57 | .57 | .56 |
| | $b_3$ | .09 | .09 | .20 | .20 | .20 | .18 |
| BSD100 | $b_2$ | .13 | .22 | .38 | .38 | .38 | .38 |
| | $b_3$ | .06 | .06 | .13 | .13 | .13 | .11 |
| Urban100 | $b_2$ | .11 | .27 | .50 | .50 | .51 | .51 |
| | $b_3$ | .02 | .03 | .14 | .14 | .15 | .13 |

priors fail on these images. The four decoders differ by at most 0.02 in either band.

**Across scale factors (E3).** We repeat the measurement at ×2, ×3, ×8 (27 cells: 3 scales × 3 bands × 3 datasets) with the same networks; ×8 is out of the training range . Pre-registered results (Table 10): the a-priori $R_b^{\text{lin}}$ ranks the measured generic-LMMSE recovery with Spearman 0.952; in 10/12 regime-C cells all four networks exceed the oracle LMMSE and differ by at most 0.03; in 9/9 regime-A cells the best network is within 0.05 of the oracle LMMSE (largest gap +0.016); generic-LMMSE recovery decreases monotonically with scale in 9/9 band–dataset pairs. The descriptive reference labels change with scale: $b_2$ is A at ×2 ($R_b^{\text{lin}} = 0.92$), B at ×3 (0.42) and C at ×8 (0.011); $b_1$ enters B at ×8 (0.79) . In all 12 C cells every network exceeds the oracle (by +0.002 to +0.244) .

**Scale collapse (L1, pre-registered).** The measurement hypothesis is that recovery is more consistent across scales when frequency is expressed relative to LR Nyquist, $u = 2s\nu$, than in absolute frequency; this comparison does not identify an observation-only causal limit. On $240^2$ crops of the first 30 images per dataset, we measure $R(u)$ in seven $u$ bins at $\times 2/\times 3/\times 4$ for LIIF, LTE and SC-INR (seed 1) and take the median range across scales, $D_{\text{in}}$ . The linear control collapses ($D_{\text{in}}$ 0.019 / 0.006 / 0.011 on Set14 / BSD100 / Urban100). Binning by $u$ instead of $\nu$ shrinks the range by an order of magnitude in 9/9 model–dataset groups (0.013–0.061 vs. 0.31–0.46). The main criterion, $D_{\text{in}} \leq 0.05$ in $\geq 8/9$ groups, fails: 6/9, with all three models on Set14 at 0.055–0.061 (BSD100 and Urban100: 0.013–0.019). This exceeds both the seed noise floor (median range across seeds 0.003–0.004) and the linear control; a post-hoc leave-one-image-out diagnostic identifies barbara as the largest contributor. Removing it gives $D_{\text{in}} = .030$–$.036$, whereas 200 fixed-seed subsets of 14 images from the first 30 BSD100 images give maxima .0378–.0392. These descriptive checks retain the original failure and do not prove a JPEG or capacity mechanism. We found no consistent deviation outside the training range (×6/8/12; $D_{\text{out}} > D_{\text{in}}$ in only 6/9), and the public

Table 10: E3 pre-registered criteria .

| Criterion | Threshold | Result |
|---|---|---|
| a. Spearman($R_b^{\text{lin}}$, generic LMMSE) | $\geq 0.8$ | 0.952 |
| b. C: nets > oracle, range $\leq 0.03$ | $\geq 80\%$ cells | 10/12 |
| c. A: best net − oracle $\leq 0.05$ | $\geq 80\%$ cells | 9/9 |
| d. generic LMMSE monotone in scale | 9/9 | 9/9 |
| e. (expl.) oracle−generic: B>A, B>C | – | .078 vs .011/.040 |

models generalise only partly (GSASR and Thera pass, ContinuousSR fails; 6/9) . Above the LR Nyquist limit, our three decoders exceed the generic linear estimate by +0.18 to +0.40 and show small differences in these evaluated cells .

**Recovered E3 implementation.** Prior simulation injects Gaussian noise with $\sigma = .002$; natural-image E3 injects none, while the GP uses assumed variance $\nu = 10^{-5}$ as regularisation. Generic-prior amplitude is scaled using HR truth variance. The LR is clipped but the linear operator is not; prior-window bands include DC whereas natural-image bands exclude it. Recovery averages RGB to grayscale, rather than using luminance. Prior sampling averages three draws, not an exact covariance trace. Saved per-image bootstrap intervals concern an unweighted mean of recoveries, not the energy-pooled table statistic. Original archives do not retain predictions, observations, band energies or true CG residuals; the present independent audit reproduces scalar records and criteria, not pixel reconstruction or numerical optimality. These differences preclude a matched-noise or deployable training-free boundary claim.

**Reading.** These evaluated networks have similar band recovery in 10/12 low-reference-recovery cells at a .03 range threshold, but only 8/12 at .02. This does not establish a universal limit on decoder design, and changing the learned prior or training recipe remains outside this comparison. The oracle-periodogram estimate is a reference rather than a certificate of optimal realised linear recovery.

## A.3 Spectral-prior effects under sparse observations (M3, M4)

**Sparse pixels.** Sampling rates 5–50%, DIV2K 0801–0812, luminance; we compare linear estimation with the generic spectrum, with the image's own spectrum, and DIP [Ulyanov et al., 2018] . The true-spectrum linear estimator beats DIP at every rate (26.39 vs. 25.19, 28.45 vs. 27.02, 32.66 vs. 30.84, 37.89 vs. 34.79 dB at 5/10/25/50%), and the generic spectrum trails it by 1.8–4.5 dB (Table 11). The space between the two linear estimators is the observed gap between these two evaluated linear references.

**1D EEG.** On 200 calibration segments of EEGMMIDB [Schalk, 2009, Schalk et al., 2004] with 10/25/50% of sam-

Table 11: Sparse-pixel reconstruction: $R_{b_2}$, $R_{b_3}$ and PSNR (dB) for generic / true-spectrum linear / DIP .

| Rate | $b_2$ | $b_3$ | PSNR |
|---|---|---|---|
| 5% | .17/.39/.16 | .10/.20/.05 | 24.59/26.39/25.19 |
| 10% | .29/.56/.34 | .14/.31/.12 | 26.13/28.45/27.02 |
| 25% | .59/.81/.69 | .29/.57/.38 | 29.27/32.66/30.84 |
| 50% | .85/.95/.89 | .57/.82/.65 | 33.37/37.89/34.79 |

Table 12: E1: 12-image mean PSNR (dB), $T = 16$ .

| Arm | $\sigma_n$=0 | .005 | .01 | .02 |
|---|---|---|---|---|
| mrgrid | 31.39 | 28.36 | 22.57 | 14.36 |
| ls_tv1e-4 | 31.16 | 27.37 | 21.24 | 13.78 |
| ls_tv1e-3 | 29.98 | 29.61 | 28.64 | 24.84 |
| ff_3 | 27.94 | 27.92 | 27.88 | 27.74 |
| ff_10 | 28.86 | 28.79 | 28.61 | 28.01 |
| gp_gauss_30 | **31.62** | 27.73 | 26.65 | 25.56 |
| gp_power_2.0 | 31.19 | 29.37 | 27.37 | 26.10 |
| gp_power_2.5 | 29.62 | 29.36 | 28.65 | 27.45 |
| gp_power_3.0 | 28.27 | 28.21 | 28.04 | 27.50 |
| gp_oracle | 29.91 | **29.70** | **29.25** | **28.35** |
| bilinear | 24.65 | 24.64 | 24.59 | 24.41 |

ples kept, the spectral prior helps most in $\theta/\alpha$ at 10–25% and almost nowhere at 50%, where most bands are already A; e.g. $\alpha$ at 10%: linear interpolation .22, population-average spectrum .30, true spectrum .47 . These are calibration-only numbers from the original report. The 760-segment test result has not been recovered and is not counted as confirmatory evidence here.

## A.4 Multi-frame SR across noise levels (E1)

**Setup.** DIV2K 0801–0812, ×4, $T = 16$ shifted frames synthesised neutrally (footprints integrated on the $512^2$ original, ground truth by $2 \times 2$ averaging), known registration, RGB, PSNR with an 8-pixel border crop; $\sigma_n \in \{0, 0.005, 0.01, 0.02\}$ . Arms: a multi-resolution grid field (mrgrid); least squares with TV (ls_tv); a SuperF-style Fourier-feature MLP (ff_$\sigma$); GPs with Gaussian spectra (gp_gauss_$\sigma$), power-law spectra (gp_power_$\alpha$) and the image's own spectrum (gp_oracle); bilinear. Paired bootstrap 95% CIs over images.

**Results.** An observed noise trend is qualitatively consistent with the scalar account, without identifying a corresponding fixed prior for the nonlinear arms: representations without spectral decay (mrgrid, weak TV) collapse under noise, losing 17 dB between $\sigma_n = 0$ and 0.02 (Table 12), an empirical deterioration rather than a demonstrated over-estimation mechanism; the value of the true spectrum over the best power-law GP grows with noise, $D$ from 0.33 to 0.85 dB, difference +0.51 [0.13, 0.88] (P3 holds); and the winner moves from a low-bias arm (gp_gauss_30) to gp_oracle (P4 holds) . At $\sigma_n = 0$ a low-bias GP beats the best Fourier-feature MLP by +2.76 [1.99, 3.56] dB.

Two pre-registered predictions fail . (P1) At $\sigma_n = 0.005$ low-bias representations no longer win: mrgrid − ff_10 = −0.43 [−1.01, 0.14] dB; the switch happens earlier than predicted. (P2) At $\sigma_n = 0.02$ the generic power-law GP does not come within 0.6 dB of the oracle (−0.85 [−1.10, −0.61]). The winner margins at $\sigma_n = 0$ and 0.005 are not significant (+0.19 and +0.10 dB, CIs include 0).

**Retrospective discrepancy diagnosis.** The pre-registered prediction used one zone label: at $T = 16$, $R^{\text{lin}}_{b_2}$ falls from 0.90 to 0.73 over the noise range (Table 17), so $b_2$ is regime A at small noise and low-bias arms were expected to win. It ignored each arm's mismatch $\kappa_b$ and ignored $b_3$. The GP-only D11 diagnostic below decomposes these discrepancies retrospectively; it does not identify the nonlinear arms' mechanisms. Two further notes. P4's switch at 0.005 partly reflects gp_oracle's numerical noise floor at $\sigma_n = 0$ (it is an oracle reference; covariance-conditioned optimality does not certify its realised per-image performance), so future switch predictions should compare practical arms only . For the non-linear arms, mrgrid and ls_tv1e-4 already lose more at $\sigma_n = 0.005$ than their noiseless lead over ff_10, which is the direct numerical reason P1 fails; their mechanism was not tested .

**Diagnostic on E1 (D11, pre-registered).** We stored per-frequency error spectra of the GP arms on the E1 images and settings (CPU; the non-linear arms mrgrid, ls_tv and ff were not measured) . At $\sigma_n = 0.005$ the excess error of gp_gauss_30 over gp_oracle splits 0.5% / 44% / 56% over $b_1/b_2/b_3$. In $b_3$ the measured oracle recovery is $R = 0.22$ (regime B) with over-estimation $\kappa = 4.5$; in $b_2$, $R = 0.80$ but $\kappa = 16.4$ (Table 13). The pre-registered rule looked only at $b_2$; $b_3$ was already in regime B, and in $b_2$ a large $\kappa$ is not free even at $R = 0.8$, since the "harmless over-estimation" of regime A is the $R \to 1$ limit. Pre-registered results: (a) the $b_3$ share of gp_gauss_30's excess is 0.556 [0.464, 0.618], above 0.5 at the point estimate but with a CI that includes it; (b) the band-wise predicted excess $\sum_b E_b\, \Delta(R_b, \kappa_b)$ ranks the measured excess over 24 arm–noise points with Spearman 0.856 [0.741, 0.918] (point threshold 0.8 holds, but the CI includes lower values); (c) it does *not* pick the best non-oracle GP arm (1/3 noise levels, threshold 2/3): the top arms are close and the band-averaged form carries a band-aggregation error (e.g. gp_power_2.0 at $\sigma_n = 0.005$: predicted $1.40 \times 10^6$, measured $3.9 \times 10^4$, unnormalised) . The concentration of excess error on $R(\nu) < 1/2$ frequencies (0.82 / 0.71 for the two wide arms vs. 0.18 for gp_power_3.0) has no discriminative value, because 78% [74%, 81%] of all frequencies satisfy

Table 13: D11: gp_gauss_30 vs. gp_oracle, E1 multi-frame, $T = 16$, $\sigma_n = 0.005$, 12-image mean .

| Band | oracle $R_b$ | $\kappa_b$ | share of excess |
|---|---|---|---|
| $b_1$ | 1.00 | 1.02 | 0.5% |
| $b_2$ | 0.80 | 16.4 | 44% |
| $b_3$ | 0.22 | 4.5 | 56% |

$R(\nu) < 1/2$ at this noise level; we do not use it as evidence. So the band-wise cost $C(Q)$ (Eq. (6)) ranks the GP arms but does not select the winner; whether its a-priori version ranks equally well has not been tested .

**Bootstrap and diagnostic boundary.** The saved D11 script used 2,000 source-image resamples. We reproduced those draws and extended the same paired procedure to the protocol's 10,000. The updated ranking CI is [.741,.918]; an independent 10,000-draw image-cluster calculation gives $b_3$ share [.464,.618]. Both condition on 12 images and fixed noise draws. D11 uses truth-derived energy, oracle recovery and spectrum ratios; its ranking is post-hoc and cannot establish a target-observation-only selector. The cropped/windowed empirical $R_b$ and band ratios also do not satisfy the exact scalar model by construction .
Non-linear arms (mrgrid, ls_tv, ff) are covered only qualitatively; $T = 4/8$ (4 images) is descriptive only.

## A.5 Inside the network: SC-INR's footprint readout (F1)

We study SC-INR's footprint readout (analytic sinc integration over the target pixel) and LTE's analogous cell input, retaining positive and negative losses. F1 asks where in the band map this component matters. We ablate it in two architecture pairs (LTE vs. LTE-NoCell; SC-INR vs. SC-INR-NoSinc) and measure the per-band loss $L = R_{\text{full}} - R_{\text{ablate}}$ on 27 confirmatory cells ($\times 2/\times 3/\times 4$ $\times$ 3 bands $\times$ 3 datasets), seed 1 . The pre-registered predictor is $g(R) = R(1-R)^2$, the small-mismatch limit of Eq. (4).

**Results (LTE pair).** $L$ is small on average in the high-reference-recovery cells: mean $L$ over regime-A cells is 0.0000, versus 0.0050 at $\times 3$ $b_2$ (regime B) . The loss concentrates in $b_3$ at $\times 2$–$\times 4$ and $b_2$ at $\times 4$, where $R_b^{\text{lin}}$ is only 0.01–0.10 (regime C) but the network recovers 0.13–0.56. The a-priori predictor fails: Spearman$(g(R_b^{\text{lin}}), L) = 0.411 < 0.5$ (standard Spearman with average ranks for ties). Using the network's own recovery raises it to 0.490, still below the threshold .

**Results (SC-INR pair).** With the matched pair (SC-INR vs. SC-INR-NoSinc, same encoder), the coarse ordering is similar but the magnitude is smaller: the A-regime criterion holds, the a-priori predictor fails (Spearman$(g(R_b^{\text{lin}}), L) = 0.400$), the network-recovery predictor also stays below threshold (0.492), and the per-cell losses of the two architectures have highly correlated ranks (Spearman 0.985) . Mean $L$ in the A cells is -0.000609, versus 0.001034 at $\times 3$ $b_2$, from the saved corrected table. All F1 numbers are seed 1 only; seeds 2/3 are evaluated under the G1 pre-registration below.

Table 14: Completed G1 confirmation; 27 aggregated cells per group.

| Group | $\rho(\epsilon_b^2, L)$ | $\rho(\epsilon_b^2 h(R), L)$ | $L_3 > L_2$ |
|---|---|---|---|
| LTE seed2 | .577 | .532 | 8/9 |
| SC seed2 | .943 | .703 | 9/9 |
| LTE seed3 | .676 | .643 | 8/9 |
| SC seed3 | .885 | .804 | 9/9 |

**A closed-form band-order predictor (G1, completed).** The box-footprint versus point-sampling quantity $\epsilon(\nu) = 1 - \text{sinc}(\nu_x)\,\text{sinc}(\nu_y)$ provides three fixed band scores (.00006, .00552, .07235). It was chosen after seed1; only seed2/3 count toward confirmation. The original protocol was recovered byte-for-byte and matched its frozen SHA256 (`548b26fe` prefix). Correctly paired HR256 evaluations contain 13 Set14, 30 BSD100 and 30 Urban100 images; $\times 3$ uses LR85. H-G1 passes in 4/4 groups (at least 3 require $\rho \geq .5$); H-G2 passes in 34/36 settings (threshold 26); H-G3 passes in 4/4: $h(R) = R(1-2R)^2$ adds no improvement within the .05 allowance. This supports coarse frequency-band ordering, not a quantitative gain mechanism. With only three distinct predictor values, tied average ranks are essential. The linearised gain account was separately rejected by L2b. Newly computed predictions were not used to tune the thresholds .

**Predicting the loss from the full model (L2, pre-registered).** One account is that the cell or footprint input tells the decoder the scale factor, so a model without it behaves like the full model averaged over the training scales. We test this without training: average the full model's output over cells corresponding to $s' \in \{1.0, 1.1, \ldots, 4.0\}$ and predict the ablation loss as $L_{\text{pred}} = R_{\text{full}} - R_{\text{marg}}$ on the 27 F1 cells, seed 1 . The ranking matches: Spearman$(L_{\text{pred}}, L) = 0.822$ (LTE) and 0.824 (SC-INR), against 0.40–0.49 for the zone predictors of F1 and $+0.147/+0.149$ above the footprint gain $\epsilon_b^2$ (0.676 on this evaluation) ; with log-uniform $s'$ it is 0.805/0.803. Two pre-registered criteria fail. The magnitude matches only at $\times 4$ and above: the median $L_{\text{pred}}/L$ is 0.59 (LTE) and 0.36 (SC-INR), outside $[0.5, 2]$ for SC-INR, because at $\times 2/\times 3$ the ablation loses more than marginalisation predicts (e.g. BSD100 $\times 3$ $b_3$: 0.038 vs. 0.008). The ablated output is closer to the marginalised than to the full output in 73.9% / 69.4% of image–scale

Table 15: L2 replication. Ratio is median $L_{\text{pred}}/L$ for $L > .005$; similarity is the fraction closer to marginal than full output.

| Group | $\rho$ | ratio | similarity | $\rho - \rho_\epsilon$ |
|---|---|---|---|---|
| LTE seed2 | .712 | .823 | .770 | +.130 |
| SC seed2 | .755 | .581 | .689 | −.142 |
| LTE seed3 | .954 | .833 | .748 | +.331 |
| SC seed3 | .700 | .337 | .653 | −.173 |

pairs (threshold 80%), rising with scale (LTE: 58% at ×2, 99% at ×8) . So the ablation loss can be predicted after the fact from the full model's cell average, consistent with the readout supplying scale information at large scale factors; at small scale factors the ablation loses more than that, and the tested scalar predictors do not establish function equivalence . This is correlational evidence; an isolated scale-information intervention has not been evaluated.

**Completed L2 seed2/3 replication.** All four frozen groups are now evaluated at HR240 (14/30/30 images), with 31 linear and 31 log-spaced cell queries. The primary 27 cells exclude ×8. Independent recalculation from per-image signal and error energies reproduces all pooled values. Under the original at-least-3/4 rule, H2a ordering passes 4/4 and H2b magnitude passes 3/4. H2c function similarity passes 0/4 (threshold .8), and H2d superiority over $\epsilon_b^2$ passes 2/4 (required improvement .05); both fail the at-least-3/4 hypothesis criterion. The advantage occurs only in LTE. H2z's cell-invariant ablation control passes 4/4 with maximum difference zero. G1 and L2 use different crops and pooling; their rows cannot be interchanged. These findings do not establish that removing the readout produces the scale-marginalised function, or that scale marginalisation is the best cross-architecture mechanistic account .

## A.6 Prospective guidance: independent calibration (G2)

**Frozen setup.** We test eight fixed GP spectra: Gaussian widths $3, 10, 30, 60, 128$ and power-law exponents $2, 2.5, 3$. The default is $1/f^2$. DIV2K training images 0001–0032 calibrate a unit stationary covariance; 0801–0812 are development images. The complete target cohort is 0813–0824 at $\sigma_n = .005, .01, .02$ (36 cases). These targets were not used to develop this selector; they are not claimed to be globally unseen in all earlier research. Matched $y = Ax+\epsilon$ uses HR256, ×4, 16 registered frames, eight footprint subsamples per axis and a 512 field. RGB PSNR uses an eight-pixel crop and unclipped predictions. This controlled geometry differs from E1's native forward model .

**Observation-only rules.** All candidates share a contrast estimate from noise-corrected observation energy and independent calibration probes. Analysis selects the minimum expected cropped MSE under the calibrated covariance, estimated by four common Gaussian probes with physical noise and the observed-mean intercept. This is a plug-in reference risk, not an unbiased target-risk estimate. Four-fold CV holds out frames by index modulo four and estimates contrast from training frames only. Each method refits its chosen prior on all frames. Each selector has 32 fits, but analysis uses full frames and CV uses 12: equal fit counts do not imply equal cost. All 36 choices and chosen outputs were SHA-frozen before scoring opened target truth. Double-precision PCG permits 4000 iterations and requires a freshly evaluated relative residual below $10^{-5}$. Earlier development failures are retained; no target cases were removed. All 36 completed with no failures; 2592 saved selection/evaluation solver certificates pass, with maximum 3134 iterations .

**Results and counterexamples.** Analysis improves the default by 1.105 dB [.726, 1.504], with no > 1 dB degradations, passing the predeclared default-improvement gate (positive lower interval bound and at most 5% severe degradations). Its gains at the three noises are $1.247, 1.124, .945$ dB. CV improves the default by 1.217 dB. The paired analysis-minus-CV gain is −.112 dB [−.313, −.002], and analysis costs 1.76 times CV. The predeclared superiority-at-no-greater-cost gate fails on both gain and cost. The frozen summarizer additionally requires zero selection failures for that gate; there are none, so this extra restriction does not change the conclusion .

Analysis chooses $1/f^{2.5}$ in 36/36 cases, whereas CV chooses exponents 2/2.5/3 in 3/27/6 cases. They hit the eight-candidate oracle in 25/36 and 34/36 cases respectively. Analysis loses to the default on image 0814 at all three noises, worst −.152 dB; its largest oracle regret is 1.248 dB on 0815 at .01 noise. A retrospective fixed-$1/f^{2.5}$ replay has identical outputs to analysis, with common preparation and final fits costing 194.6 s. This algebraic/cache identity is an interpretation of the completed results, not an added confirmatory baseline or a new default chosen before this test. The experiment does not isolate a benefit of target-specific analytical selection .

**Full cost and practical reading.** Analysis and CV require $5.065 \times 10^{12}$ and $3.222 \times 10^{12}$ interpolated points. Peak recorded allocated memory is .488 GB. Calibration takes 3.0 s separately; the actual resource ledger records 14,346.6 s of GPU-phase invocation wall time, including the experiment's selection and oracle scoring, within the 24-hour budget. These timings depend on the shared-server load and are not a benchmark of exclusive-device speed. Independent CPU audits reproduce statistics and

Table 16: Complete G2 cohort. Gain is relative to fixed $1/f^2$; 95% intervals use 10,000 bootstrap draws of 12 image clusters, averaging three noises within each image. Regret is against the best of the eight fixed candidates scored after freezing. Costs include common contrast preparation, all selector fits and the chosen final fit, once per deployed policy; shared fits are not added twice to actual experimental spending. Calibration overhead is reported separately.

| Rule | Mean gain [95% CI], dB | Regret mean/median/p90, dB | Worst gain, dB | $>1$ dB loss | Cost, s |
|---|---|---|---|---|---|
| Default | 0 | 1.219 / 1.064 / 1.801 | 0 | 0/36 | 160.5 |
| Analysis | 1.105 [.726, 1.504] | .114 / 0 / .134 | −.152 | 0/36 | 8466.8 |
| Four-fold CV | 1.217 [.754, 1.796] | .002 / 0 / 0 | 0 | 0/36 | 4821.7 |

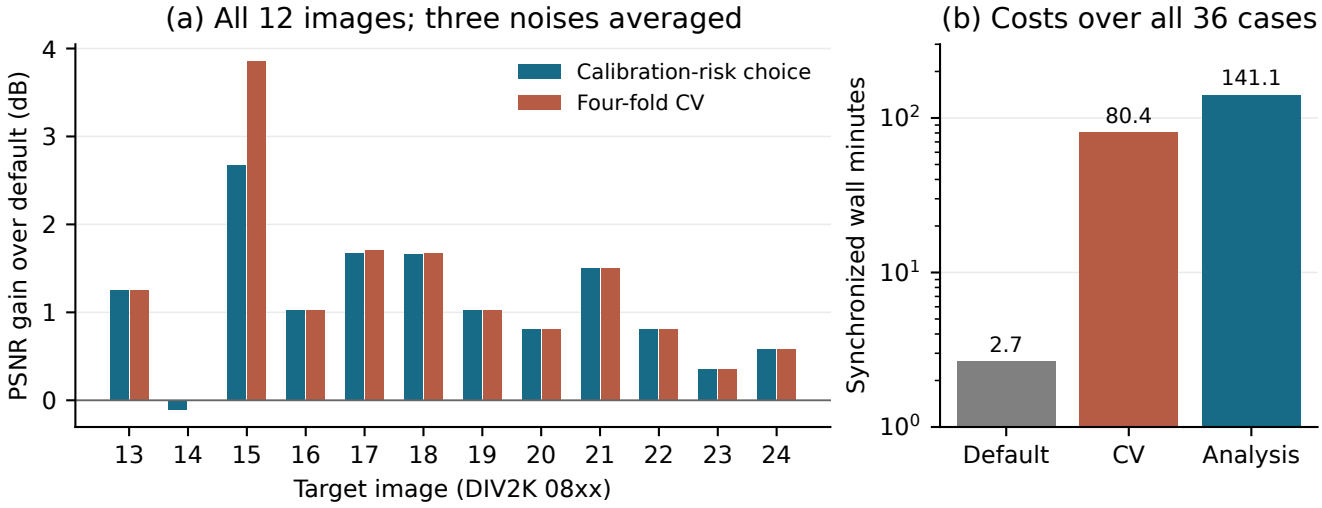



Figure 3: G2 complete-cohort gains and costs. (a) Every target image is shown; averaging three noise settings does not create three independent image replicates. (b) Synchronized wall time of common preparation, selection and final fits on the same A800, on a log axis; this is not GPU busy time. The analysis rule chooses power exponent 2.5 in all 36 cases, so its gain does not establish instance-specific adaptation.

costs; all 78 cached chosen predictions were re-scored from RGB pixels. Uncached candidate predictions were not retained, so their oracle scores were audited from saved MSE and convergence records, without an independent GP rerun. In this task, calibrated analysis finds a better prior than the specified default, but frame-holdout CV adapts more effectively and costs less. We therefore retain the analysis as a conditional diagnostic, not a demonstrated cheap replacement for CV or a neural-architecture selector .

# B Proof of Proposition 3

With $R = p/(p+q)$ the matched gain $R$ attains error $p(1-R)$. The gain $g = \kappa p/(\kappa p + q) = \kappa R/(\kappa R + 1 - R)$ gives error $(1-g)^2 p + g^2 q$. Using $q/p = (1-R)/R$ and $D = \kappa R + 1 - R$,

$$\frac{(1-g)^2 p + g^2 q}{p} = \frac{(1-R)\big[(1-R) + \kappa^2 R\big]}{D^2}.$$

Subtracting $(1-R)$ and expanding $D^2$, the numerator becomes $(1-R)\,R(1-R)(\kappa-1)^2$, which is Eq. (4). For Cor. 1, $\Delta(R,\kappa)/\Delta(R,1/\kappa) = \big((1-R)\kappa + R\big)^2/\big(1-R+\kappa R\big)^2$, which exceeds 1 for $\kappa > 1$ iff $(\kappa-1)(1-2R) > 0$, i.e. iff $R < 1/2$ (also checked numerically).

# C Reproducibility

Original sources, protocols and results retain their server directory names in the collected evidence. The updated `SOURCES.md` maps each claim to original or new output. New G1 and L2 runs invoke original frozen evaluation code and preserve old results. The G1 protocol matches SHA256 prefix `548b26fe`; the full hash is recorded in `SOURCES.md`. Independent L2 checking reconstructs pooled recoverability from saved per-image signal/error arrays; G1 checking validates saved pooled numbers and coverage, but per-image energy weights were not saved by its original script. Exact machine-readable numerical checks accompany the evidence.

# D Historical reference map

These prior simulations retain the original numerical values and fixed labels for source traceability. They do not use the natural-image E3 observation law: the latter has no injected noise and uses a different numerical variance. The labels are descriptive, with threshold sensitivity unverified.

# E Matrix optimality and aggregation counterexamples

Let $e_P = x - H_P y$. Since $\mathbb{E}[e_P y^\top] = 0$, expanding $x - Hy = e_P + (H_P - H)y$ cancels both cross terms. Multiplication by an arbitrary fixed $W$ and taking traces gives the identity in §3. Gaussianity is unnecessary for this second-moment calculation.

Even equal-energy modes with identical $\kappa = 4$ demonstrate that band plug-in costs have no uniform error direction. For $R = (.1, .9)$ the average exact cost is .218639; for $R = (.01, .99)$ it is .041601. Both have average $R = .5$, whose plug-in cost is .18: an underestimate in the first example and an overestimate in the second. These examples do not use non-diagonal sampling, so aggregation alone suffices. For fixed finite $\kappa > 0$ the cost vanishes as $R \to 0$ or 1, but the low-$R$ limit is not uniform in $\kappa$: $R = .01$, $\kappa = 100$ gives $\Delta = 24.256863$. The scalar identity was independently checked on a $101 \times 121$ grid with maximum algebraic residual $7.2 \times 10^{-15}$ .

# F Exploratory same-observation linear controls

This development diagnostic separates a model's own uncertainty from actual reconstruction error. It uses two already-seen images, DIV2K 0801/0802, one Fourier-feature seed and one paired noise draw; it is neither a confirmatory comparison nor a new selection method. A central native $128 \times 128$ crop is average-pooled to $64 \times 64$. Two point-sampling frames use vertical stride four and horizontal phases $(0, 0)$, $(0, 1)$ or $(0, 2)$, each with 512 noisy RGB observations and noise variance $\nu = 10^{-4}$. Only horizontal phases change; this matched point operator differs from E1's native integrated footprints.

All 24 linear predictions are fixed before truth-based scoring: two images, three geometries, two scales $\sigma \in \{3, 10\}$ and two kernels. The RBF is $k_\sigma(u, v) = \exp[-2\pi^2\sigma^2\|u - v\|^2]$. The finite Fourier kernel uses the original network's 256 frozen Gaussian frequencies, with $\phi(u) = [\cos(2\pi Bu), \sin(2\pi Bu)]/\sqrt{256}$ and $k(u, v) = \phi(u)^\top\phi(v)$. Both have unit diagonal, fixed mean $m = .5$ and ridge parameter $\nu$, solved in float64 by Cholesky. No bandwidth, amplitude or regularisation search is performed. Saved nonlinear FF outputs are compared without retraining. Fixed Fourier ridge is not the MLP's full NTK or learned kernel [Tancik et al., 2020]; its mean, regularisation and optimisation also differ from those of the FF network. The noise generator resets to the same seed for each geometry, giving common random numbers across geometries.

Let $T = K_{to}(K_{oo} + \nu I)^{-1}$ for the $P$ scored target pixels. Conditioning on a fixed clean RGB image, the exact raw-output risk over fresh observation noise is

$$\frac{\|T(X_o - m) - (X_t - m)\|_F^2}{3P} + \frac{\nu\|T\|_F^2}{P}.$$

Truth is used only to score the bias term after the complete prediction cohort is hash-frozen. For the saved noise draw, actual raw MSE further includes its realised noise term and signed bias–noise cross term. These identities do not apply after clamping, and are not observation-only risk estimators. In contrast, $\mathrm{diag}(K_{tt} - TK_{ot})$ is uncertainty under the candidate's own Gaussian prior. Different kernels specify different target laws, so their own uncertainties are not a common actual-risk comparison.

Table 18 has three opposing finite-Fourier comparisons. At $\sigma = 10$, phase $(0, 2)$ has conditional expected *raw* MSE 1.901/2.473 on 0801/0802, predominantly bias; clipping must not conceal this failure. Within each fixed kernel/scale, model posterior variance orders the three phases as 02 < 01 < 00, while the conditional noise propagation term orders them as 00 < 02 < 01. Total conditional raw-risk ordering agrees with model variance in only five of eight image/kernel/ scale combinations. Lower self-prior uncertainty therefore does not certify lower fixed-image error.

On the discrete $56 \times 56$ target crop, phases $(0, 1)$ and $(0, 2)$ have exactly the same nearest-observation distance histogram. Continuous periodic maximum gaps three/two alone cannot explain their different errors, especially for these nonperiodic kernels and nonnested observation sets. Covariance-parameter identifiability and reconstruction error are different objects; this diagnostic supplies no general transfer theorem. Joint correlations, prior mismatch and estimator behaviour require separate analysis rather than a nearest-distance or rank-only mechanism.

All 24 controls finish with no numerical failure in 17.39 seconds of shared CPU wall time (four threads, nice 10, 600-second absolute budget). The complete prediction manifest precedes scoring; all saved outputs, linear responses, residuals and failures are retained. Independent LU recomputation of all 24 controls and primal/dual checking of all 12 Fourier controls agree with predictions within $2.51 \times 10^{-11}$. These audits validate the calculations, not natural-image prior validity, practical guidance or publication novelty .

As a common-law diagnostic, the same 12 distinct saved linear responses are evaluated on 32 separate calibration crops, DIV2K 0001–0032, with identical preprocessing,

Table 17: Historical reference simulations: $R_b^{\mathrm{lin}}$ per band under the generic $1/f^2$ prior, assumed spectrum and simulated observations; not a natural-image bound . Labels: A $\geq 0.8$, B, C $< 0.1$.

| Task | $\sigma_n$ | $b_1$ | $b_2$ | $b_3$ |
|---|---|---|---|---|
| Single image ×2 | .002 / .01 | 1.00 / 1.00 | .92 / .90 | .09 / .08 |
| Single image ×4 | .002 / .01 | .98 / .98 | .09 / .08 | .01 / .01 |
| Single image ×8 | .002 / .01 | .79 / .77 | .01 / .01 | .01 / .01 |
| Sparse 5% | .002 | .70 | .09 | .03 |
| Sparse 10% | .002 | .81 | .20 | .07 |
| Sparse 25% | .002 | .92 | .47 | .20 |
| Sparse 50% | .002 | .98 | .76 | .46 |
| Multi-frame ×4, $T$=4 | .002→.02 | ≈1.0 | .73→.51 | .12→.03 |
| Multi-frame ×4, $T$=8 | .002→.02 | ≈1.0 | .83→.62 | .27→.08 |
| Multi-frame ×4, $T$=16 | .002→.02 | ≈1.0 | .90→.73 | .41→.15 |

| Estimator | $\sigma$ | 0801 | 0802 |
|---|---|---|---|
| RBF | 3 | +.763 | +.552 |
| RBF | 10 | +1.792 | +1.908 |
| Fixed Fourier ridge | 3 | +.480 | −1.125 |
| Fixed Fourier ridge | 10 | −.647 | −.664 |
| Trained FF | 3 | +.633 | +.443 |
| Trained FF | 10 | +.951 | +.919 |

Table 18: Development difference in clamped PSNR (dB), phase $(0, 2)$ minus $(0, 1)$, on the identical shave-four region. All opposing signs are retained; these two images and one seed do not establish a population ordering or a nonlinear mechanism.

mean, noise variance and target region. Averaging the preceding conditional-risk formula over these complete RGB images defines a finite empirical evaluation law; no Gaussian or stationarity assumption or population confidence interval is attached to it. All 384 image/operator risks and eight static recommendations are hash-frozen before consulting the two development images. This is a retrospective diagnostic designed after their earlier outcomes, not prospective evidence. Calibration chooses RBF $\sigma = 3$ within every geometry, and phase $(0, 2)$ globally, agreeing with the conditional-risk optimum on both development images. Across the overlapping eight choice groups per image, 15 of 16 recommendations match their oracle; the exception is Fourier ridge $\sigma = 3$ on 0801, with raw-risk regret .001037. Nevertheless, 15 of 132 pairwise comparisons have opposing signs. These dependent comparisons do not establish a general transfer rate. The calibration recommendation within each geometry coincides with the fixed RBF $\sigma = 3$ baseline; there is no image-adaptive gain. The complete CPU diagnostic takes 8.23 seconds without solving new systems or retraining .

# G Ambiguity, selection and unrestricted reconstruction

This elementary control illustrates classical robust decision principles, not a new minimax theorem [Kitagawa, 2011]. On eight periodic sites let $Q_k(i, j) = \frac{1}{4}\cos[2\pi k(i - j)/8]$, $D = Q_1 - Q_3$, and $C_\pm = I \pm \theta D$, with known $0 < \theta < 1$, unknown sign, zero mean and known noise variance $\nu > 0$. The priors are positive definite. Write $a = 1 + \nu$ and let $A_{01}$ and $A_{02}$ select sites $(0, 4, 1, 5)$ and $(0, 4, 2, 6)$ respectively. All risks below are total eight-coordinate squared error; divide by eight for per-coordinate MSE.

For $A = A_{02}$, $ADA^\top = 0$ and $\mathrm{tr}(AD^2A^\top) = 2$. Both worlds therefore have exactly the same observed Gaussian law, $N(0, aI)$, and posterior means $m_\pm = A^\top y/a \pm \theta DA^\top y/a$. Regret subtracts each world's own known-sign Bayes risk. Their midpoint is an admissible mixture estimate. Posterior orthogonality and the parallelogram identity give

$$\frac{\mathrm{Reg}_+(g) + \mathrm{Reg}_-(g)}{2} = \mathbb{E}\|g - A^\top y/a\|^2 + \frac{2\theta^2}{a}.$$

The midpoint attains both regrets equally, hence the unrestricted minimax regret is $2\theta^2/a$ and total minimax risk is $8 - 4/a$. Choosing only between $m_+$ and $m_-$ instead gives minimax regret $4\theta^2/a$ when randomisation is allowed; fixed wrong selection costs $8\theta^2/a$. Candidate-selection failure must not be promoted to the same lower bound for all reconstruction methods.

The known-spectrum oracle favours $A_{02}$ by the positive risk gap

$$\delta = R^*_{01} - R^*_{02} = \frac{\theta^2(1 + 2\nu - \theta^2)}{a(a^2 - \theta^2/2)} < \frac{\theta^2}{a}.$$

For $K$ independent calibration draws observed through the chosen geometry, followed by a fresh evaluation draw, both designs pay $4(K + 1)$ measurements. Geometry 02 cannot learn the sign for any $K$. Geometry 01 has distinct laws; calibration-only likelihood selection has

error at most $\frac{1}{2}\exp[-K\theta^2/(4a^2)]$ and wrong-rule cost $\Gamma = 8a\theta^2(a^2 - a + 1/2)/(a^2 - \theta^2/2)^2$. It beats geometry 02's unrestricted worst-case total risk whenever $\Gamma\exp[-K\theta^2/(4a^2)]/2 < 2\theta^2/a - \delta$. This is a sufficient finite threshold, not a sharp transition or a new general sampling law. Free full-resolution calibration or spectrum labels would invalidate the information restriction. Prediction versus covariance-learning design is already studied [Zimmerman, 2006]; the present two-point illustration supplies no natural-image or INR guarantee .

# H Finite-sample Gaussian confidence-set study

The completed frozen 8-point stationary Gaussian study contains 1920 cases under known zero mean, known noise and fixed trace-eight spectra. It uses independent same-law draws for selection and fresh-draw population risk for evaluation. All selections were hash-frozen before truth scoring. Of 1920 cases, 1887 confidence sets were nonempty, 33 empty and none reported numerical failure; the true spectrum lay in 1841 sets. Only 197 cases (10.26% of the full cohort) yielded a recommendation certificate; no incorrect recommendation was observed among those 197. This does not mean zero error over the full cohort or safety on an adaptively reused single image. All 11322 valid LP endpoints and 384 diagnostic endpoints were independently recomputed. Strong empirical-Bayes performance and low certification coverage preclude a practical-advantage claim. The construction uses classical covariance estimation, partial identification and contrast-design ideas; it is not claimed as a new general sampling theorem. Protocols, null cases, raw outputs and full costs remain in the source ledger.